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Effects of electronic medical record in a Chinese hospital: A time series study

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ABSTRACT

Background: Electronic medical record (EMR) systems have potential to improve care quality and efficiency. Yet little is known about the effects of EMR in Chinese hospitals.
Objective: An inpatient EMR system was implemented in a large Chinese hospital. The objective of this research was to evaluate the impact of the EMR system on efficiency, quality, and cost of inpatient care in the hospital.
Methods: A five-year (2005–2009) longitudinal dataset with monthly data involving 251 physicians and 298,760 patient visits was created by querying the hospital's central database. Four key outcomes were examined: length of stay, infection rate, mortality rate, and cost per patient stay. Interrupted time series analysis was used to evaluate changes in level and slope for each outcome variable.
Results: Length of stay grew at 0.027 bed-days per month in the pre-EMR period and declined at 0.043 bed-days per month in the post-EMR period. Infection rate rose at 0.036 infections per 100 patients per month in the pre-EMR period and declined at 0.062 infections per 100 patients per month in the post-EMR period. Mortality rate grew at 0.048 deaths per 1000 patients per month in the pre-EMR period and decreased at 0.005 deaths per 1000 patients per month in the post-EMR period. Cost per patient stay declined at 33 RMB per month in the pre-EMR period and increased at 16 RMB per month in the post-EMR period.
Conclusions: The EMR was associated with the declining patterns of length of stay, infection rate, and mortality rate. It had no positive effect on patient costs.
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1. Introduction

Electronic medical record (EMR) systems have drawn global attention from healthcare researchers, practitioners, and policy makers due to their potential of containing healthcare costs, enhancing efficiency, and improving patient safety and care quality [1–4]. Believing in its transformational potential, researchers enthusiastically advocate the use of EMR [5]. Hillestad et al. [1] suggested that the annual savings in the United States due to EMR could be over $81 billion. Recently, the United States government implemented the HITECH Act to provide financial incentives to spur adoption and meaningful use of EMR [6]. A total of $17 billion was designated for this purpose and an eligible provider who can demonstrate meaningful use of EMR will receive $41,000 in the first five years after adopting EMR. In addition to the US, many countries such as Australia, New Zealand, and England have been pushing for

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the adoption of EMR [5]. Given the ongoing passion on EMR in the global scope, it is imperative to empirically demonstrate the positive impact of EMR on health care.

Hundreds of studies have evaluated EMR's impact on health care. Since several systematic reviews on EMR evaluation have been published, we can rely on these published reviews to get a glimpse of the existing knowledge on EMR. A review of 257 articles shows that EMR can improve quality (increased adherence to guideline-based care, enhanced surveillance and monitoring, decreased medication errors) and efficiency (decreased utilization) of health care [7]. Chaudhry et al. [7] also find mixed results on time utilization of EMR – some find that EMR can save physician time, while others find the contrary. A review of 67 articles concerned with hospitalized patients [8] shows that computerized physician order entry (CPOE), a component of the EMR system, is effective in increasing guideline adherence, generating appropriate alerts, reducing costs, improving operational efficiency, and reducing medication errors. Another review of 30 articles on EMR [9] reveals that EMR has both positive and negative effects on primary care outpatient practices, and EMR can only marginally improve operational efficiency and health-related outcomes. Moreover, some studies showed that EMR was not associated with quality improvement in ambulatory care [10,11]. A study even found that CPOE facilitated medication error risks [12]. However, physicians seem to have a favorable perception of EMR. A national survey [13] shows that physicians reported positive effects of EMR in improving care quality and were highly satisfied with EMR. A recent study finds that physician productivity can be greatly improved after the implementation of EMR [14]. Although not often examined, EMR's capability storing and transferring digital medical data can facilitate intra- and inter-organizational communication and clinical research, thus improving efficiency and quality indirectly. Overall, no consensus has been achieved regarding the impact of EMR systems on health care and more empirical research is needed to shed light in this area.

While research on EMR evaluation abounds, most studies were carried out in developed countries. Little is known about EMR's impacts on health care in the emerging economies. Holroyd-Leduc et al. [9] calls for more research on EMR assessment in developing countries so that the impact of EMR can be better understood in the global context. Responding to this call, the objective of this study is to study the impact of an EMR system on three health care outcomes: efficiency, quality, and costs in a large university-affiliated hospital in China. Employing a time series study method, we intend to test whether the EMR implementation is associated with reduced length of stay, infection rate, mortality rate, and patient costs.

This study will contribute to the medical informatics literature in three ways. First, it provides additional empirical evidence for EMR's effects on efficiency, quality, and costs of health care. By analyzing longitudinal archival data from a large hospital using segmented regression models, we were able to show how the change dynamics of outcome variables differed before and after the EMR implementation. Second, due to practical constraints, the pre-post design represents the majority of previous EMR evaluations although it has many limitations [7–9]. In the real world, it is not always possible to conduct a controlled evaluation due to resource and policy constraints. A major limitation of the pre-post design is that it compares the outcome variable at two time points and cannot reveal the trend of change over time [15]. Our study analyzes the trends of outcome variables over 60 months, thus providing more insights than pre-post evaluations. Finally, little research has been conducted to evaluate EMR systems in China where health IT has been greatly advanced [16–18]. This study could deepen our understanding about EMR given its international context.

2. Methods

2.1. Setting

Jiangyin People's Hospital (JPH) is the site of this study. As one of China's top 100 hospitals, JPH has 1150 beds, more than 1304 employees, and 429 physicians. The hospital has more than 20 years of information technology (IT) experiences and is a pioneer of using health information technology in China. It was accredited as one of the ten Modernized Hospitals in the Jiangsu province. Its network infrastructure connects more than 50 functional software systems that involve most hospital units. It has implemented comprehensive administrative information systems to automate hospital management tasks such as cost management, resource management, medication management, office automation, accounting, physician performance assessment, and patient registration. Clinical information systems were implemented in many units such as nursing stations, pharmacy, laboratories, radiology, outpatient departments, and inpatient wards. JPH's strategic goal is to use IT to lead organizational change and process redesign, thereby creating business value and improving care efficiency and quality.

Since the beginning of 2006, JPH started implementing an EMR system for inpatient care. The system was expected to improve quality and efficiency and reduce patient costs. For this study, we selected length of stay as an efficiency indicator, infection rate and mortality rate as quality indicators, and patient costs as a costs indicator. In JPH, the clinical performance of each physician was monitored and electronically stored. Merit-based salaries for the physicians were associated with their clinical performance. This management approach allowed us to collect data on the selected outcomes longitudinally and carry out detailed statistical analysis. Although each inpatient ward had a slight different timeline for the EMR implementation, all of the physicians started using the EMR in January, 2007. Hence, this month was set as the EMR go-live date.

2.2. The EMR system

JPH's EMR was used to support all of the clinical and administrative processes involved in a patient's hospital stay. As shown in Fig. 1, the system consisted of several key subsystems: computerized physician order entry (CPOE), nursing information system (NIS), pharmacy information system (PIS), laboratory information system (LIS), picture archiving and communication system (PACS), and clinical data repository (CDR). All of the patient data and clinical information are

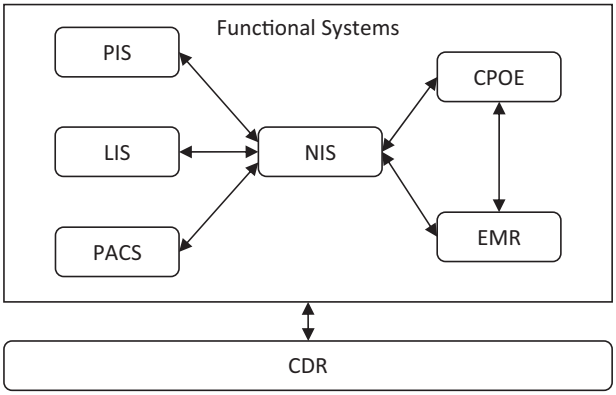

Fig. 1 – Information systems implemented in JPH.

saved in the CDR and shared by all of the subsystems that are seamlessly connected via the hospital intranet. The CPOE allows physicians and residents to conduct diagnosis, order laboratory and radiology tests, prescribe medications, and provide dictations. The NIS functionalities include patient registration, admission/transfer/discharge, bed assignment, care planning and reporting, medication administration, surgery application, and billing. Nurses can view physician orders and dictations which are fed from the CPOE into the NIS. The laboratory and medication orders will be double-checked in the NIS by nurses to eliminate computer input errors before being submitted to laboratories or pharmacies. The orders will be processed by relevant units such as the hospital laboratory and pharmacy and results will be sent back to the EMR after being checked again at the NIS.

Clinical decision support capabilities have been integrated into the EMR. First, a number of common order sets were created in the CPOE so that physicians can order a group of medications more efficiently and accurately. Second, the EMR consists of a database of the Chinese national pharmacopeia. Users can use an interface in the CPOE to search for a drug and get detailed drug information. Third, the CPOE automatically generate drug safety alerts such as side effects, drug–drug interaction, drug–food interaction, and contraindications to physicians when they input orders into the system. Finally, the system allows physicians to search and view specific clinical information to facilitate their decision making. For example, they can view detailed nursing reports, medication history, periodical vital sign changes, and medical images before deciding whether a patient should be discharged. According to the HIMSS Analytics EMR Adoption Model [19], JPH's EMR is at Stage 3.

2.3. *Measurements*

Four outcome variables were measured: length of stay (LOS), infection rate, mortality rate, and patient costs. LOS is commonly investigated as an indicator of efficient use of hospital resources. Many studies reported reduced LOS as a result of EMR [7,15,20], while some find no evidence that EMR can reduce LOS [21]. Infection rate and mortality rate were selected as indicators for health care quality because they are part of the Agency for Healthcare Research and Quality (AHRQ) quality indicators [22]. The AHRQ inpatient quality indicators include mortality rates for various conditions and procedures, and the AHRQ patient safety indicators include infection rate due to medical care as an important measure [22]. In addition, prior research has found healthcare IT is associated with reduced mortality rate and infection rate [21,23,24]. Patient costs were selected because one of the major benefits of using EMR is its capability of cutting costs [1]. While some studies showed that EMR can reduce costs [25], systematic reviews suggested that limited evidence exists on the relationship between EMR and costs [7]. Hence, by assessing these four outcomes, we intended to explore the effects of EMR on efficiency, quality, and cost of health care in JPH.

These four outcome variables are aggregated monthly data at the hospital level. Length of stay was measured by the average hospital bed-days of the patients discharged in a given month. Infection rate was measured by the number of infections per 100 hospitalized patients that occurred during hospitalization in a given month. Similarly, mortality rate referred to the number of deaths per 1000 hospitalized patients that occurred during hospitalization in a given month. Cost per patient stay was calculated by using the overall inpatient cost divided by the number of admissions during a given month. Raw data were retrieved from the JPH central database and values of the outcome variables were calculated.

2.4. *Procedure*

Approval for this study was obtained from the University & Medical Center Institutional Review Board of East Carolina University and the Department of Research and Education of JPH. A time series study involving 251 physicians in JPH's inpatient wards was carried out. Monthly data from January, 2005 to December, 2009 were collected. Although it was desirable to have a controlled experiment, it was impossible for this study because we could not find a similar hospital that was willing to disclose data on the four outcome variables to us. In China, the clinical information such as mortality and infection rate and patient costs is highly sensitive and confidential, and hospitals have no responsibility of sharing such information with the public. We could not have an internal control group either, because all of the inpatient wards in JPH implemented EMR. Hence, the time series design was practically the best option for us.

2.5. *Data collection*

Data related to 251 physicians and 298,760 patient visits was obtained. The average age of the physicians was 31.86 (SD = 6.92). The majority of physicians (68%) were male. The data for length of stay, infection rate, mortality rate, and patient costs were based on these patients. Since the patient data were aggregated, the patients' demographic information was unknown.

The data were retrieved from the JPH clinical data repository by using structured query language (SQL). A set of SQL queries were developed to select and aggregate records from

various database tables based on specific search criteria on clinical team and time. Since the four outcome variables are based on data stored in different tables, the "from" clause of the SQL queries had to join multiple tables to combine data sources. Math functions and "group by" commands were used to transform daily and individual patient level data into monthly aggregates. After preparation, all of the data related to the 60 months between 2005 and 2009 were saved in an Excel file after necessary cleansing, translation, and transformation.

2.6. *Statistical methods*

Following Wagner et al. [26], we analyzed the 60-month data as an interrupted time series. This type of analysis is commonly used to analyze the effects of interventions in the situation where a control group is difficult or impossible to find [27–30]. In this model, the variation within the data was partitioned into three components to provide independent tests for (1) the slope in scores for the pre-EMR period, (2) the change in level for the post-EMR period, accounting for the pre-EMR trend, and (3) the change in slope from pre-EMR to post-EMR.

In the model estimation, each outcome variable's growth over time was linear before and after the EMR implementation. Three predictors were included in the linear regression model: MONTH, EMR, and POSTEMR. MONTH was coded as sequential numbers so that 1 represents January, 2005 and 60 represents December, 2009. EMR was a dummy variable that was set as 0 before the EMR go-live date (January, 2007) and 1 thereafter. POSTEMR was created to measure months of EMR using experiences after the go-live date. The score of POSTEMR was 0 for all months before January, 2007, and incremented by 1 in each month afterwards. MONTH was used to estimate the outcome variable's slope in scores in the pre-EMR period (test 1). EMR was used to estimate the change in levels for the post-EMR period (test 2). POSTEMR was used to estimate the difference in slope from the pre-EMR to the post-EMR period.

To control for autocorrelations in our monthly data, we employed stepwise autoregression to analyze the data. Thirteen autoregressive lags were estimated to account for seasonality and other possible secular trends. Significant lags were kept in the model. The maximum likelihood method was applied to fit the regression models and estimate the standard errors of regression coefficients. A p value of less than 0.05 was considered to indicate statistical significance. SAS 9.2 was used for the analysis.

3. Results

3.1. *Length of stay*

As shown in Table 1 and Fig. 2, LOS had been increasing before the EMR go-live month (January, 2007). The rate of increase was 0.027 bed-days per month from January, 2005 through December, 2006 (SD = 0.011, $p = 0.02$). In January, 2007, after the EMR was used by all of the 251 physicians, LOS rose by 0.295 bed-days, but not significantly, higher than expected, as compared with the trend before the go-live month ($p = 0.13$). Subsequently, the rate of increase dropped below the pre-EMR period and the direction of the slope reversed from positive to negative. LOS started decreasing at the rate of 0.043 bed-days per month from January, 2007 through December, 2009 ($p < 0.001$).

3.2. *Infection rate*

Infection rate was increasing before the EMR go-live month (Fig. 2 and Table 1). The rate of increase was 0.036 infections per 100 patients from January, 2005 through December, 2006 ($p < 0.001$). In January, 2007, infection rate rose by 0.244 bed-days, but this increase was not significant ($p = 0.13$). After January, 2007, the post-EMR trend reversed from increasing to decreasing. Infection rate dropped at the rate of 0.062 infections per 100 patients per month in the post-EMR period ($p < 0.001$).

3.3. *Mortality rate*

Mortality rate was increasing before the EMR go-live month (Fig. 2 and Table 1). The rate of increase was 0.048 deaths per 100 patients from January, 2005 through December, 2006 ($p = 0.001$), and after the EMR go-live month, mortality rate dropped by 1.039 deaths per 1000 patients ($p < 0.001$). In the post-EMR period, the trend reversed and mortality rate started declining. From January, 2007 through December, 2009 mortality rate was decreasing at the rate of 0.005 deaths per 1000 patients per month ($p = 0.001$).

3.4. *Patient cost*

As shown in Table 1 and Fig. 2, cost per patient was decreasing at the rate of 33 RMB per month in the pre-EMR period ($p = 0.002$). After the EMR go-live month, cost per patient grew by 317 RMB, but this increase was not significant ($p = 0.08$). In the post-EMR period, the trend reversed and cost per patient started increasing. From January, 2007 through December, 2009, cost per patient was increasing at the rate of 16 RMB per month ($p < 0.001$).

4. Discussion

As expected, EMR showed positive effects on LOS, infection rate, and mortality rate. The growth trajectories of LOS, infection rate, and mortality rate changed from a pre-EMR ascending trend to a post-EMR descending trend. Although JPH's EMR system is a specific implementation, it shares many common features other EMR systems have. For example, it electronically store patients' clinical data to give physicians and nurses timely access. The CPOE component enables physicians to order tests and medications easily and the alert function can reduce order entry errors. Laboratory and radiology results are made available to physicians immediately after the tests by LIS and PACS systems, saving valuable time during medical procedures. The connection between PIS and NIS allows pharmacists and nurses to confirm that correct prescriptions are filled and administered to patients. It is possible that these features contributed to quality and efficiency

Table 1 – Summary of interrupted time series analysis.

Variables	Slope: January 2005 to December 2006 (standard error)	*p* value	Change in level. Predicted score of January 2007 based on post-EMR trend minus predicted score based on pre-EMR trend (standard error)	*p* value	Change in slope from January 2005–December 2006 to January 2007–December 2009 (standard error)	*p* value
Length of stay	0.027 (0.011)	0.02	0.295 (0.194)	0.13	−0.070 (0.012)	<0.001
Infection rate (infections per 100 patients)	0.036 (0.016)	<0.001	0.244 (0.242)	0.32	−0.098 (0.021)	<0.001
Mortality rate (deaths per 1000 patients)	0.048 (0.014)	0.001	−1.039 (0.253)	<0.001	−0.053 (0.016)	0.001
Cost per patient (in 1000 RMB)	−0.033 (0.010)	0.002	0.317 (0.181)	0.08	0.049 (0.012)	<0.001

of medical services at JPH, thus reducing LOS, infection rate, and mortality rate.

Surprisingly, although the pre-EMR slope of cost per patient was descending, the slope became ascending in the post-EMR period. In China, inpatient costs have been increasing at an annual rate of 6–10% in recent years [31]. The cost escalation results from a complex set of policy, economic, technology, and healthcare factors and is unlikely to be addressed by EMR alone. First, the current health policy in China ties physician incomes to hospital revenues, which gave rise to physicians' irrational prescribing behaviors and increasing healthcare costs [32–34]. Second, while experiencing an economic boom, China suffers from inflation that contributed to the increasing healthcare costs [35]. Third, technology advancement leads to new drugs, devices, and procedures, and the use of these innovations inevitably drove up costs [36]. Finally, with lifestyle changing and population aging, chronic diseases become increasingly prevalent in China [37], adding additional burden to the healthcare system. Our results suggest that the EMR implementation in JPH had no effect on reducing patient costs. It should be noted that the ease of placing orders using CPOE might have implicitly made physicians to write more orders. More research is needed to examine whether CPOE could lead to excessive order writing.

The literature has provided mixed findings about the effects of EMR systems on efficiency, quality, and cost of health care [7,10,11]. Some prior studies that employed a cross-sectional design or compared absolute levels of outcome variables at two time points could not reveal the outcomes' growth dynamics over time. Such studies may not be able to fully examine the potential effects of EMR. Interrupted time series analysis has advantages over cross-sectional analyses and pre-post comparisons. It is able to evaluate not only the change of absolute levels at the time of intervention, but also the change of the slope after the intervention, controlling for the preexisting level and slope. This type of time series design can reduce threats to internal validity such as history and maturity by making multiple outcome measurements both before and after the EMR implementation [26].

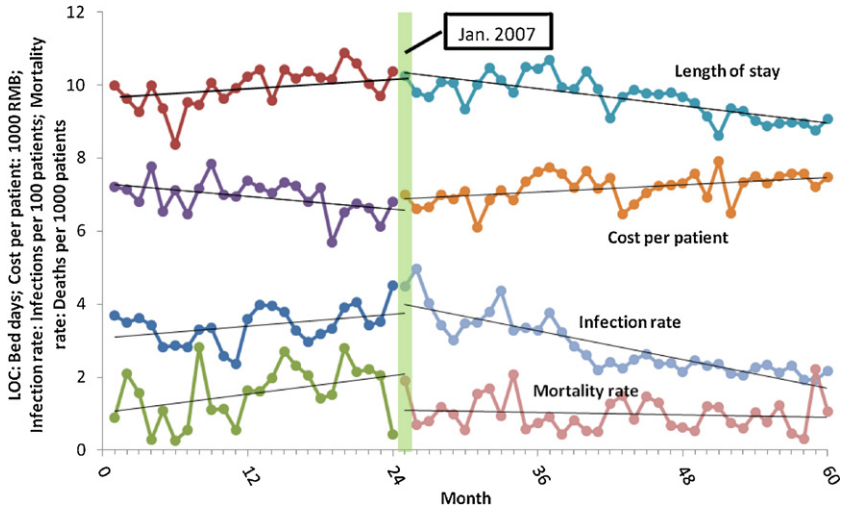

Fig. 2 – Effects of EMR on length of stay, infection rate, mortaility rate, and patient cost.

In addition, research on information systems has shown that users' productivity and efficiency could decrease at the early stage of the system implementation because users need time to learn how to use the system and adjust to the internal process logic embedded in the system [38–40]. This study finds that LOS and infection rate became greater in the EMR go-live month (Fig. 2). This is possibly because while the physicians were trying to integrate EMR into their medical practices the efficiency and quality of their services were negatively affected. Fig. 2 also indicates that in the post-EMR period, LOS and infection rate kept decreasing and dropped lower than the pre-EMR level, which suggests that the benefits of EMR started to accrue after physicians overcome the learning curve.

Our study has several limitations. First, constrained by data availability, we could not measure more outcome variables to comprehensively reflect the broad scope of quality, efficiency, and cost of health care. It is unknown whether the EMR implementation is associated with changes in other outcomes. Second, a time series design was utilized in this study without a control group. The trend changes of LOS, infection rate, and mortality rate might be due to other events rather than the EMR implementation. Although we examined the 60-month period carefully and did not find any events that could have contributed to the changes, this study does not provide empirical evidence to infer causation from EMR to the trend changes of LOS, infection rate, and mortality rate in JPH. Third, we did not examine disease-specific LOS, infection rate, and mortality rate. Yet, these outcomes might be influenced by different factors such as disease surveillance, treatment adherence, and natural environment for each specific disease. Our findings should be interpreted with these limitations in mind.

In summary, this study analyzed the changes in LOS, infection rate, mortality rate, and cost per patient stay before and after the implementation of an EMR system in a large teaching hospital in China during the 60-month period from 2005 to 2009. Based on interrupted time series analyses, we conclude that EMR was associated with positive changes of LOS, infection rate, and mortality rate. After the EMR implementation, the direction of the slopes of LOS, infection rate, and mortality rate changed from ascending to descending. In contrast, the slope of cost per patient stay changed from descending to ascending. Therefore, the EMR implemented at JPH seemed to have positive impacts on LOS, infection rate and mortality rate and no impact on patient costs.

Author contributions

Yajiong Xue and Huigang Liang contributed for the research design, data acquisition and analysis, manuscript writing. Xiaocheng Wu contributed for the data acquisition and analysis, manuscript writing. Hai Gong, Bin Li and Yuxia Zhang contributed for the data acquisition, manuscript writing.

Conflicts of interest

None.

Summary points

What was already known on the topic

- EMR has potential to improve care quality
- EMR has potential to improve care efficiency

What this study added to our knowledge

- There is no improvement in terms of length of stay and infection rate immediately after the EMR implementation.
- Longitudinally, EMR is associated with the declining trend of length of stay, infection rate, and mortality rate.
- We find no evidence that EMR can reduce patient cost.

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